Describe your previous research experience or equivalent experience on complex projects, including the level of independence, while working as a member of a research/project team.

As an undergraduate research assistant in the Vermaas lab at the MSU-DOE Plant Research Laboratory, I work on computational biophysics projects focused on simulating biological systems and characterizing their behavior. I independently set up simulation systems using VMD and CHARMM-GUI, run molecular dynamics (MD) simulations on high-performance computing clusters like MSU HPCC and NCSA DeltaAI using NAMD, and build analysis pipelines using Python and Tcl. I periodically incorporate feedback from my postdoc and PI to refine my approaches. I have spent the last year working on a project to characterize water transport modulation mechanisms in a capped aquaporin channel, which yields insights into the mechanisms behind plant drought tolerance and responses. I set up and run multiple MD simulations exposing the channel to different osmotic gradients, and analyze the resulting trajectories by computing metrics such as pore radius profiles, water permeation rates, and ion distributions. I am also responsible for producing figures for presentations for meetings with experimental collaborators, as well as for conferences such as ACS Fall 2025 and the First Great Lakes Plant Science Conference. Over the last four months, I have also worked on simulating the effect of alcohols on bacterial membrane properties, with the broader goal of informing bioengineering efforts towards the design of tolerant strains for industrial biofuel production. I have conducted end-to-end MD simulations of bacterial membranes with varying alcohol concentrations, and am in the process of writing a first-author manuscript on the insights gained from these simulations.

As an Honors Research Scholar, I am conducting research on quantum subgroup simulators under the guidance of Professor Ryan LaRose. My work involves developing methods for efficiently simulating quantum circuits on classical hardware, which is essential for the development and evaluation of larger-scale quantum algorithms that cannot yet be run on existing quantum hardware. Over the last three months, I have been exploring literature on group-theoretic and Lie-theoretic approaches to unitary subgroup membership, designing and implementing optimization routines to determine optimized gatesets for circuit simulation, and creating circuit rewriting algorithms to reduce the comuptational cost of simulation. I discuss my progress weekly with Professor LaRose to reevaluate future directions and ensure alignment with larger research goals.